

$$\begin{aligned}
& \wedge router_ct = 0 \\
& \wedge shard_ct = 0 \\
& \wedge tx_status = \\
& \quad [t \in AllTxns \mapsto \text{IF } t = InitTx \text{ THEN "Committed" ELSE "Inactive"}] \\
& \wedge tx_snapshot = [t \in AllTxns \mapsto 0] \\
& \wedge tx_checkset = [t \in AllTxns \mapsto \{\}] \\
& \wedge tx_buffer = [t \in AllTxns \mapsto [k \in Keys \mapsto \langle \rangle]] \\
& \wedge tx_commit_time = [t \in AllTxns \mapsto 0] \\
& \wedge store = \\
& \quad [k \in Keys \mapsto \{[val \mapsto InitVal, cts \mapsto 0, writer \mapsto InitTx]\}] \\
& \wedge pc = [t \in TxnIds \mapsto 1] \\
& \wedge op_seq = [t \in TxnIds \mapsto \langle \rangle] \\
\\
StartTx(t) & \triangleq \\
& \wedge tx_status[t] = \text{"Inactive"} \\
& \wedge \text{LET } sess \triangleq Workload[t].session \\
& \quad lvl \triangleq Workload[t].level \\
& \text{IN IF } lvl \in \{\text{"PC"}, \text{"SI"}, \text{"SER"}\} \\
& \quad \text{THEN } \wedge tx_snapshot' = [tx_snapshot \text{ EXCEPT } ![t] = shard_ct] \\
& \quad \wedge \text{UNCHANGED } router_ct \\
& \quad \text{ELSE IF } lvl \in \{\text{"CC"}, \text{"PSI"}\} \\
& \quad \text{THEN } \wedge tx_snapshot' = [tx_snapshot \text{ EXCEPT } ![t] = client_dep[sess]] \\
& \quad \wedge \text{UNCHANGED } router_ct \\
& \quad \text{ELSE } \boxed{RA} \\
& \quad \wedge tx_snapshot' = \\
& \quad \quad [tx_snapshot \text{ EXCEPT } ![t] = client_last_commit[sess]] \\
& \quad \wedge \text{UNCHANGED } router_ct \\
& \wedge tx_status' = [tx_status \text{ EXCEPT } ![t] = \text{"Active"}] \\
& \wedge \text{UNCHANGED } \langle client_dep, \\
& \quad client_last_commit, \\
& \quad shard_ct, \\
& \quad tx_checkset, \\
& \quad tx_buffer, \\
& \quad tx_commit_time, \\
& \quad pc, \\
& \quad store, \\
& \quad op_seq \\
& \rangle \\
\\
MaxVersion(versions, max_cts) & \triangleq \\
& \text{LET } valid \triangleq \{v \in versions : v.cts \leq max_cts\} \\
& \text{IN CHOOSE } v \in valid : \forall v2 \in valid : v.cts \geq v2.cts \\
\\
ExecuteOp(t) & \triangleq
\end{aligned}$$